Ex 5	Analysis of Prompting Techniques in LLM Reasoning

Learning Objective:

Develop a Python program to analyze how different prompting strategies—Zero-shot, Few-shot, and Chain-of-Thought (CoT)—affect the reasoning capabilities of Large Language Models (LLMs). The program will also identify patterns in logical fallacies or hallucinations using benchmark datasets.

Steps:

1. Import Required Libraries: OpenAI API or Hugging Face Transformers or Google Gemini API and Additional libraries: datasets, pandas, matplotlib
2. Configure API Keys
   - Set your OpenAI and Gemini API keys securely (do not share).
3. Load Dataset
   - Use either SQuAD (QA reasoning) or GSM8K (math reasoning).
4. Initialize Models
5. Define Prompting Strategies
   - Zero-shot Prompt, Few shot prompt, Chain-of-Thought Prompt
6. Create Functions to Query Models
7. Generate Outputs
   - Apply all prompting techniques on the same dataset questions.
8. Display Results of Zero-shot Prompt, Few shot prompt, Chain-of-Thought Prompt

Code:

```
import warnings, matplotlib.pyplot as plt
from datasets import load_dataset
from transformers import pipeline
warnings.filterwarnings("ignore")
print("Loading GPT-2 model...")
gen = pipeline("text-generation", model="gpt2", pad_token_id=50256)
print("✓ Model ready\n")
samples = load_dataset("gsm8k", "main", split="test").select(range(3))
```

```
def zero_shot(q): return f"Answer this math question:\n{q}\nAnswer:"
def few_shot(q):  return ("Q: A shop has 5 apples and gets 3 boxes of 4. Total?\n"
                    "A: 5 + 3×4 = 17 apples.\n" f"Q: {q}\nA:")
def cot(q):       return ("Solve step by step.\n"
                    "Q: 3 baskets × 6 oranges - 4=?\n"
                    "Step 1: 3×6=18. Step 2: 18-4=14. Answer:14\n"
                    f"Q: {q}\nStep 1:")
def ask(prompt):
    out = gen(prompt, max_new_tokens=80, truncation=True)[0]["generated_text"]
    return out[len(prompt):].strip()
questions = [(s["question"], s["answer"].split("####")[-1].strip()) for s in samples]
scores = {"Zero-shot": 0, "Few-shot": 0, "CoT": 0}
print("=" * 60)
print("          STRATEGY 1: ZERO-SHOT PROMPTING")
print("  (No examples given — model answers directly)")
print("=" * 60)
zs_results = []
for q, ref in questions:
    out = ask(zero_shot(q))
    if ref in out: scores["Zero-shot"] += 1
    zs_results.append(out)
    print(f"\nQ : {q[:80]}...")
    print(f"Ref: {ref}")
    print(f"Ans: {out[:200]}")
    print("-" * 60)
print("\n" + "=" * 60)
print("          STRATEGY 2: FEW-SHOT PROMPTING")
print("  (2 examples provided before the question)")
print("=" * 60)
fs_results = []
for q, ref in questions:
    out = ask(few_shot(q))
    if ref in out: scores["Few-shot"] += 1
    fs_results.append(out)
```

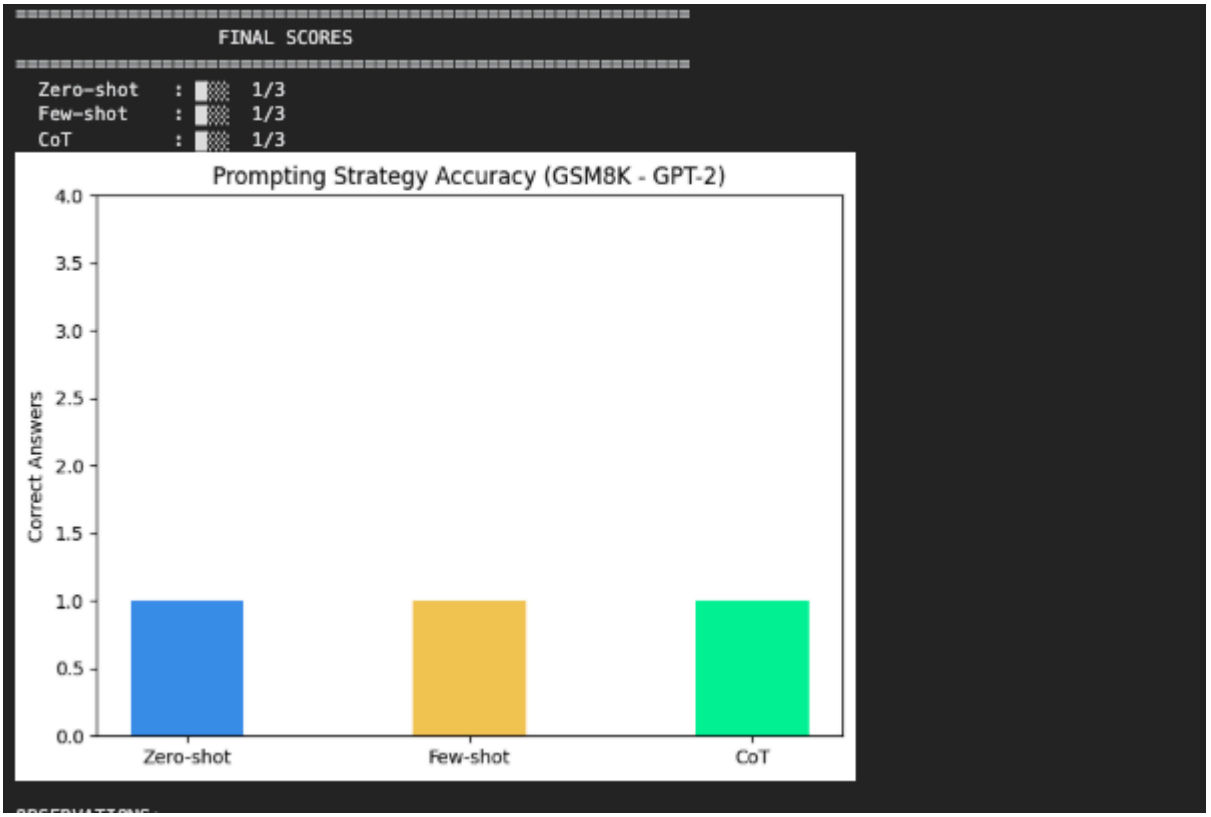
```
    print(f"\nQ : {q[:80]}...")
    print(f"Ref: {ref}")
    print(f"Ans: {out[:200]}")
    print("-" * 60)
print("\n" + "=" * 60)
print("         STRATEGY 3: CHAIN-OF-THOUGHT (CoT)")
print("  (Step-by-step reasoning guided)")
print("=" * 60)
cot_results = []
for q, ref in questions:
    out = ask(cot(q))
    if ref in out: scores["CoT"] += 1
    cot_results.append(out)
    print(f"\nQ : {q[:80]}...")
    print(f"Ref: {ref}")
    print(f"Ans: {out[:200]}")
    print("-" * 60)
print("\n" + "=" * 60)
print("                FINAL SCORES")
print("=" * 60)
for k, v in scores.items():
    bar = "█" * v + "░" * (3 - v)
    print(f"  {k:12}: {bar}  {v}/3")
plt.bar(scores.keys(), scores.values(), color=["#4e9af1","#f1c94e","#4ef19a"], width=0.4)
plt.title("Prompting Strategy Accuracy (GSM8K - GPT-2)")
plt.ylabel("Correct Answers"); plt.ylim(0, 4); plt.tight_layout()
plt.savefig("prompting_results.png", dpi=120); plt.show()
print("""
OBSERVATIONS:
- Zero-shot : No examples, model guesses directly. Lower accuracy.
- Few-shot  : Examples improve structure and format of answers.
- CoT       : Step-by-step reasoning reduces errors. Best for math.
✓ Chart saved as prompting_results.png
""")
```

Output:

```
============================================================
          STRATEGY 1: ZERO-SHOT PROMPTING
  (No examples given — model answers directly)
============================================================
Both `max_new_tokens` (=80) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information.

Q : Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning an...
Ref: 18
Ans: It's $5.92.
Feb 5, 2006 at 1:27 p.m.
Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the
------------------------------------------------------------
Both `max_new_tokens` (=80) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information.

Q : A robe takes 2 bolts of blue fiber and half that much white fiber.  How many bol...
Ref: 3
Ans: I have a 3.5 bolt. How many bolts in total does it take?
Answer:  I have a 4.5 bolt.  How many bolts in total does it take?
Answer:  I have a 5.25 Bolt. How many bolts in total does it take?
Answer:
------------------------------------------------------------
Both `max_new_tokens` (=80) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information.

Q : Josh decides to try flipping a house.  He buys a house for $80,000 and then puts...
Ref: 70000
Ans: $50,000.
Answer: $40,000.  He has to pay $15,000 in repairs and $10,000 in taxes.
Answer:
So in the end, Josh gets $40,000 in profit from the home.  This profit is split between Josh and his wife.  H
------------------------------------------------------------
```

```
============================================================
          STRATEGY 2: FEW-SHOT PROMPTING
  (2 examples provided before the question)
============================================================
Both `max_new_tokens` (=80) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information.

Q : Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning an...
Ref: 18
Ans: $2.50/day.
Q: Is Janet the only duck that has two sets of eyes?
A: No.
Q: Janet's duck is like a duck. She has a set of eyes that are slightly raised and have a set of eyes that are slightly raised. T
------------------------------------------------------------
Both `max_new_tokens` (=80) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information.

Q : A robe takes 2 bolts of blue fiber and half that much white fiber.  How many bol...
Ref: 3
Ans: 1 + 1×4 = 5 bolts.
Q: A shop has 5 apples and gets 3 boxes of 4. Total?
A: 5 + 3×4 = 17 apples.
Q: A robe takes 2 bolts of blue fiber and half that much white fiber. (Theoretically this makes a robe t
------------------------------------------------------------
Both `max_new_tokens` (=80) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information.

Q : Josh decides to try flipping a house.  He buys a house for $80,000 and then puts...
Ref: 70000
Ans: Josh made $60,000.  As of this writing, he makes $4,200.
Q: Josh is trying to sell the house to a friend.  The friend doesn't notice.  He just buys a house for $80,000 and then moves out.  Josh is sti
------------------------------------------------------------
```

```
============================================================
          STRATEGY 3: CHAIN-OF-THOUGHT (CoT)
  (Step-by-step reasoning guided)
============================================================
Both `max_new_tokens` (=80) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information.

Q : Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning an...
Ref: 18
Ans: 7 eggs per day. Step 2: 6 eggs per day. Answer:6
Q: 1-2 eggs per day. Step 3: 3 eggs per day. Answer:3
Q: 5-8 eggs per day and eggs at least 1/3 off. Answer:5
Step 1: 6 eggs per day. Answer:6
Step 2:
------------------------------------------------------------
Both `max_new_tokens` (=80) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information.

Q : A robe takes 2 bolts of blue fiber and half that much white fiber.  How many bol...
Ref: 3
Ans: Step 2:
Step 3:
Step 4:
Step 5:
Step 6:
Step 7:
Step 8:
Step 9:
Q: What's the difference between a hat and a scarf?
Step 1:
Step 2:
Step 3:
Step 4:
Step 5:
------------------------------------------------------------

Q : Josh decides to try flipping a house.  He buys a house for $80,000 and then puts...
Ref: 70000
Ans: $80,000
Step 2: $50,000
Step 3: $150,000
Step 4: $200,000
Step 5: $300,000
Step 6: $400,000
Step 7: $500,000 - Step 8: $600,000
Step 9: $700,000 - Step 10: $800,000
```



Learning Outcome:

Students will:

- Understand how Zero-shot, Few-shot, and CoT prompting influence LLM reasoning.
- Identify patterns in hallucinations and logical fallacies.
- Gain experience working with benchmark datasets like GSM8K or SQuAD.